

Task 3 : Login System using Express.js, MongoDB (Mongoose), Sessions, and Class

Create a simple login system using Express.js, MongoDB (Mongoose), express-session, and JavaScript Class.

Task Description:

Build a login system where users can:

1. Register a new account
2. Login using username and password
3. Access a protected dashboard
4. Logout from the system

Requirements:

1. Database:

- Create MongoDB database named: studentDB
- Create collection named: users
- Each user must have:
 - username
 - password

2. User Class:

Create a class called User with methods:

- register()
- login()

Example:

```
class User {
  constructor(username, password) {
    this.username = username;
    this.password = password;
  }

  async register() {}
  async login() {}
}
```

3. Express Routes:

Register Route:

POST /register

- Save user in MongoDB

Login Route:

POST /login

- Check user from MongoDB
- If correct, create session

Example:

```
req.session.user = username;
```

Dashboard Route (Protected):

GET /dashboard

- Only logged-in users can access
- Show message: Welcome username

Logout Route:

GET /logout

- Destroy session

4. Middleware:

Create authentication middleware to protect dashboard route.

Expected Output:

Register:

User registered successfully

Login:

Login successful

Dashboard:

Welcome username

Logout:

Logout successful

Requirements:

- Express.js
- MongoDB
- Mongoose
- express-session
- JavaScript Class

Submission Requirements:

Students must submit:

- server.js file
- User class file
- MongoDB connection code
- Working routes

Other Submission Requirement

- Git Committees

GitHub repository link

- AI generated code with a comparison report

- Example commit structure:

```
git init
```

```
git add .
```

```
git commit -m "Initial project setup with Express server"
```

```
git commit -m "Added MongoDB connection using Mongoose"
```

```
git commit -m "Created User class with register method"
```

```
git commit -m "Implemented login route with session"
```

```
git commit -m "Added dashboard protected route"
```

```
git commit -m "Implemented logout functionality"
```

-Report Requirements:

Students must generate code using an AI tool (such as ChatGPT) and compare it with their own written code.

Create a report document containing:

Feature	Student Code	AI Code
Structure	Simple / Advanced	Simple / Advanced
Readability	Easy / Medium / Hard	Easy / Medium / Hard
Security	Basic / Good	Basic / Good
Session Handling	Yes / No	Yes / No